

PROJECT-3

E-commerce Application

1] DOCKER BUILD

```
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$ nano Dockerfile
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$ nano .dockerignore
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$ docker build -t devops-build .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
             Install the buildx component to build images with BuildKit:
             https://docs.docker.com/go/buildx/

Sending build context to Docker daemon   2.62MB
Step 1/5 : FROM nginx:alpine
----> 04da2b0513cd
Step 2/5 : RUN rm -rf /usr/share/nginx/html/*
----> Running in ld6adc09ee31
----> Removed intermediate container ld6adc09ee31
----> 01b92c209a61
Step 3/5 : COPY build/ /usr/share/nginx/html/
----> 9063f6002c35
Step 4/5 : EXPOSE 80
----> Running in 1179010f2b89
----> Removed intermediate container 1179010f2b89
----> f185bdce66fb
Step 5/5 : CMD ["nginx", "-g", "daemon off;"]
----> Running in 1b1423732f6b
----> Removed intermediate container 1b1423732f6b
----> 5c60dea0470e
Successfully built 5c60dea0470e
Successfully tagged devops-build:latest
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$
```

2] DOCKER-COMPOSE,BUILD.SH & DEPLOY FILE CREATION

```
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$ nano docker-compose.yml
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$ nano .dockerignore
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$ nano build.sh
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$ nano deploy.sh
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$ sudo chmod +x build.sh
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$ sudo chmod +x deploy.sh
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$ ./build.sh
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
             Install the buildx component to build images with BuildKit:
             https://docs.docker.com/go/buildx/

Sending build context to Docker daemon   2.62MB
Step 1/5 : FROM nginx:alpine
----> 04da2b0513cd
Step 2/5 : RUN rm -rf /usr/share/nginx/html/*
----> Using cache
----> 01b92c209a61
Step 3/5 : COPY build/ /usr/share/nginx/html/
----> Using cache
----> 9063f6002c35
Step 4/5 : EXPOSE 80
----> Using cache
----> f185bdce66fb
Step 5/5 : CMD ["nginx", "-g", "daemon off;"]
----> Using cache
----> 5c60dea0470e
Successfully built 5c60dea0470e
Successfully tagged devops-build:latest
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$ ./deploy.sh
WARN[0000] /home/ubuntu/project-3/E-commerce-Application/docker-compose.yml: 'version' is obsolete
+! Running 1/1
```

3] DOCKER-COMPOSE-BUILD

```
ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=ap-south-1&connType=standard&instanceId=i...
aws Search [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) Account ID: 0459-9814-6012 sudharsan

--> 01b92c209a61
Step 3/5 : COPY build/ /usr/share/nginx/html/
--> Using cache
--> 9063f6002c35
Step 4/5 : EXPOSE 80
--> Using cache
--> f185bdce66fb
Step 5/5 : CMD ["nginx", "-g", "daemon off;"]
--> Using cache
--> 5c60dea0470e
Successfully built 5c60dea0470e
Successfully tagged devops-build:latest
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$ ./deploy.sh
WARN[0000] /home/ubuntu/project-3/E-commerce-Application/docker-compose.yml: `version` is obsolete
[+] Running 1/1
  ✓ Network e-commerce-application default Removed 0.1s
WARN[0000] /home/ubuntu/project-3/E-commerce-Application/docker-compose.yml: `version` is obsolete
[+] Pulling 1/1
  ✗ web Error 2.4s
Error response from daemon: pull access denied for devops-build, repository does not exist or may require 'docker login': denied: requested access to t
he resource is denied
WARN[0000] /home/ubuntu/project-3/E-commerce-Application/docker-compose.yml: `version` is obsolete
[+] Running 1/2
  ✓ Network e-commerce-application default Created 0.8s
  ✓ Container e-commerce-application-web-1 Started 0.5s
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$ docker ps
CONTAINER ID   IMAGE               COMMAND                  CREATED        STATUS        PORTS                               NAMES
fcfd04ac172b   devops-build:latest "/docker-entrypoint..." About a minute ago Up About a minute 0.0.0.0:80->80/tcp, [::]:80->80/tcp e-commerce-application-web-1
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$
```

4] DEV REPO CREATION DOCKERHUB

hub.docker.com/repository/create?namespace=sudharsanps115

hub Explore My Hub Search Docker Hub CtrlK

sudharsanps115 Docker Personal

Repositories

Hardened Images

Collaborations

Settings

Default privacy

Notifications

Billing

Usage

Pulls

Storage

Repositories / Create

Create repository

Repository Name *
dev

Short description
repo for project-3

A short description to identify your repository. If the repository is public, this description is used to index your content on Docker Hub and in search engines, and is visible to users in search results.

Visibility

Using 0 of 1 private repositories. [Get more](#)

☒ Public Appears in Docker Hub search results

☐ Private Only visible to you

[Cancel](#) [Create](#)

Pushing images

You can push a new image to this repository using the CLI:

```
docker tag local-image:tagname new-repo:tagname
docker push new-repo:tagname
```

Make sure to replace tagname with your desired image repository tag.

5] PROD REPO CREATION DOCKERHUB

hub Explore My Hub

Search Docker Hub CtrlK

Using 0 of 1 private repositories. [Get more](#)

Create repository

Repository Name *
prod

Short description
repo for project-3

A short description to identify your repository. If the repository is public, this description is used to index your content on Docker Hub and in search engines, and is visible to users in search results.

Visibility

Using 0 of 1 private repositories. [Get more](#)

☐ Public Appears in Docker Hub search results

☒ Private Only visible to you

Cancel Create

Pushing images

You can push a new image to this repository using the CLI:

```
docker tag local-image:tagname new-repo:tagname
docker push new-repo:tagname
```

Make sure to replace `tagname` with your desired image repository tag.

6] DOCKER LOGIN

```
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$ docker login

USING WEB-BASED LOGIN

Info -> To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: XSMF-GDNZ
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...

WARNING! Your credentials are stored unencrypted in '/home/ubuntu/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
devops-build  latest   5c60dea0470e   3 hours ago    56.4MB
nginx         alpine   04da2b0513cd   3 weeks ago    53.7MB
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$
```

7] DOCKERHUB PUSH

```
ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=ap-south-1&connType=standard&instanceId=i...
aws [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) Account ID: 0459-9814-6012 sudharsan

ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$ docker tag devops-build:latest sudharsanps115/dev:latest
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$ docker images
REPOSITORY          TAG         IMAGE ID      CREATED       SIZE
devops-build         latest      5c60dea0470e  3 hours ago  56.4MB
sudharsanps115/dev   latest      5c60dea0470e  3 hours ago  56.4MB
dockerhub-sudharsanps115/dev latest      5c60dea0470e  3 hours ago  56.4MB
nginx                alpine     04da2b0513cd  3 weeks ago  53.7MB
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$ docker push sudharsanps115/dev:latest
The push refers to repository [docker.io/sudharsanps115/dev]
7ca7d3d57c38: Pushed
6aef49cdac4f: Pushed
e6fe11fa5b7f: Pushed
67ea0b046e7d: Pushed
ed5fa8595c7a: Pushed
8ae63eb1f31f: Pushed
b3e3d1bbb64d: Pushed
48078b7e3000: Pushed
fd1e40d7f74b: Pushed
7bb20cf5ef67: Pushed
latest: digest: sha256:4851d6e148c0c972135b8838368ab11e866fb7bceab20d7fbb038552f2c63593 size: 2406
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$
```

8] PUSHED IMAGES IN DEV9] REPO

hub Explore My Hub

Search Docker Hub CtrlK

sudharsanps115 Docker Personal

Repositories

Hardened Images

Collaborations

Settings

Default privacy

Notifications

Billing

Usage

Pulls

Storage

Repositories / dev / General

sudharsanps115/dev

Last pushed 1 minute ago · Repository size: 22.6 MB · ☆0 · ↓0

repo for project-3

Add a category

General Tags Image Management BETA Collaborators Webhooks Settings

Tags

This repository contains 0 tag(s).

Tag	OS	Type	Pulled	Pushed
latest		Image	less than 1 day	1 minute

See all

Repository overview

An overview describes what your image does and how to run it. It displays in the public view of your repository once you have pushed some content.

Using 1 of 1 private repositories.

Docker commands

To push a new tag to this repository:

docker push sudharsanps115/dev:tagname

buildcloud

Build with Docker Build Cloud

Accelerate image build times with access to cloud-based builders and shared cache.

Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated per-organization isolation.

Get faster builds through shared caching across your team, native multi-platform support, and encrypted data transfer - all without managing infrastructure.

Go to Docker Build Cloud

9] JENKINS INSTALLATION

```
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-01-08 10:22:22 UTC; 3min 37s ago
     Main PID: 21093 (java)
        Tasks: 43 (limit: 2204)
      Memory: 517.2M (peak: 534.6M)
         CPU: 22.484s
       CGroup: /system.slice/jenkins.service
               └─21093 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Jan 08 10:22:16 ip-172-31-10-230 jenkins[21093]: [LF]> This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Jan 08 10:22:16 ip-172-31-10-230 jenkins[21093]: [LF]>
Jan 08 10:22:16 ip-172-31-10-230 jenkins[21093]: [LF]> *****
Jan 08 10:22:16 ip-172-31-10-230 jenkins[21093]: [LF]> *****
Jan 08 10:22:16 ip-172-31-10-230 jenkins[21093]: [LF]> *****
Jan 08 10:22:22 ip-172-31-10-230 jenkins[21093]: 2026-01-08 10:22:22.665+0000 [id=39] INFO jenkins.InitReactorRunner$1#onAttained: Compil
Jan 08 10:22:22 ip-172-31-10-230 jenkins[21093]: 2026-01-08 10:22:22.691+0000 [id=30] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is
Jan 08 10:22:22 ip-172-31-10-230 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Jan 08 10:22:22 ip-172-31-10-230 jenkins[21093]: 2026-01-08 10:22:24.233+0000 [id=56] INFO h.m.DownloadService$Downloadable#load: Obtain
Jan 08 10:22:24 ip-172-31-10-230 jenkins[21093]: 2026-01-08 10:22:24.236+0000 [id=56] INFO hudson.util.Retrier#start: Performed the acti

ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
3307abb3a36c4787b1841e009fdeccae
ubuntu@ip-172-31-10-230:~/project-3/E-commerce-Application$
```

10] JENKINS DASHBOARD

**Jenkins**

[+ New Item](#)

[Build History](#)

Build Queue

No builds in the queue.

Build Executor Status

0/2

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job

Set up a distributed build

Set up an agent

Configure a cloud

Learn more about distributed builds

REST API

Jenkins 2.528.3

11] JENKINS CRED

Jenkins / Manage Jenkins / Credentials

Credentials

AKIAQVNNUAHOOWHU4EYA/*****	System - aws - aws-cred
sudharsanps115/*****	System - dockerhub - DOCKERHUB

Stores scoped to Jenkins

System

Domains: Global, aws, dockerhub

11] JENKINS PIPELINE OUT

Jenkins / e-commerce application / #6 / Console Output

Console Output

Download Copy View as plain text

```
Started by user sud
Obtained E-commerce-Application/Jenkinsfile from git https://github.com/sudharsanps115/FINAL-PROJECT
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/e-commerce application
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/e-commerce application/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/sudharsanps115/FINAL-PROJECT # timeout=10
Fetching upstream changes from https://github.com/sudharsanps115/FINAL-PROJECT
> git --version # timeout=10
> git --version # 'git version 2.43.0'
> git fetch --tags --force --progress -- https://github.com/sudharsanps115/FINAL-PROJECT +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/dev^{commit} # timeout=10
Checking out Revision bde31f88752e2739674c60a5f53c8a423033dc6f (refs/remotes/origin/dev)
> git config core.sparsecheckout # timeout=10
> git checkout -f bde31f88752e2739674c60a5f53c8a423033dc6f # timeout=10
Commit message: "Update Jenkinsfile"
> git rev-list --no-walk df8912317924fc03c2d38b84128330fb9b37962 # timeout=10
[Pipeline] }
```

13] JENKINS PIPELINE CONFIG 1

The screenshot shows the Jenkins Configuration page for a job named 'e-commerce application'. The 'General' tab is selected in the left sidebar. The 'Enabled' toggle is turned on. The 'Description' field is empty. Below it, there are several checkboxes: 'Discard old builds' (unchecked), 'Do not allow concurrent builds' (checked), 'Abort previous builds' (unchecked), 'Do not allow the pipeline to resume if the controller restarts' (unchecked), 'GitHub project' (unchecked), 'Pipeline speed/durability override' (unchecked), 'Preserve stashes from completed builds' (unchecked), and 'This project is parameterised' (unchecked). At the bottom, there are 'Save' and 'Apply' buttons.

Configure

- General
- Triggers
- Pipeline
- Advanced

General Enabled ☒

Description

Plain text [Preview](#)

☐ Discard old builds ?

☒ Do not allow concurrent builds

☐ Abort previous builds ?

☐ Do not allow the pipeline to resume if the controller restarts

☐ GitHub project

☐ Pipeline speed/durability override ?

☐ Preserve stashes from completed builds ?

☐ This project is parameterised ?

[Save](#) [Apply](#)

14] JENKINS PIPELINE CONFIG 2

The screenshot shows the Jenkins Configuration page for the same job, but with the 'Triggers' tab selected. The 'Build periodically' checkbox is unchecked. The 'GitHub hook trigger for GITScm polling' checkbox is checked. The 'Poll SCM' checkbox is unchecked. The 'Trigger builds remotely (e.g., from scripts)' checkbox is unchecked. Below the triggers, there is a 'Pipeline' section with a 'Definition' dropdown set to 'Pipeline script from SCM'. Under 'SCM', the 'Git' option is selected. In the 'Repositories' section, the 'Repository URL' is set to 'https://github.com/sudharsanps115/FINAL-PROJECT'. At the bottom, there are 'Save' and 'Apply' buttons.

Configure

- General
- Triggers
- Pipeline
- Advanced

☐ Build periodically ?

☒ GitHub hook trigger for GITScm polling ?

☐ Poll SCM ?

☐ Trigger builds remotely (e.g., from scripts) ?

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM ?

Git

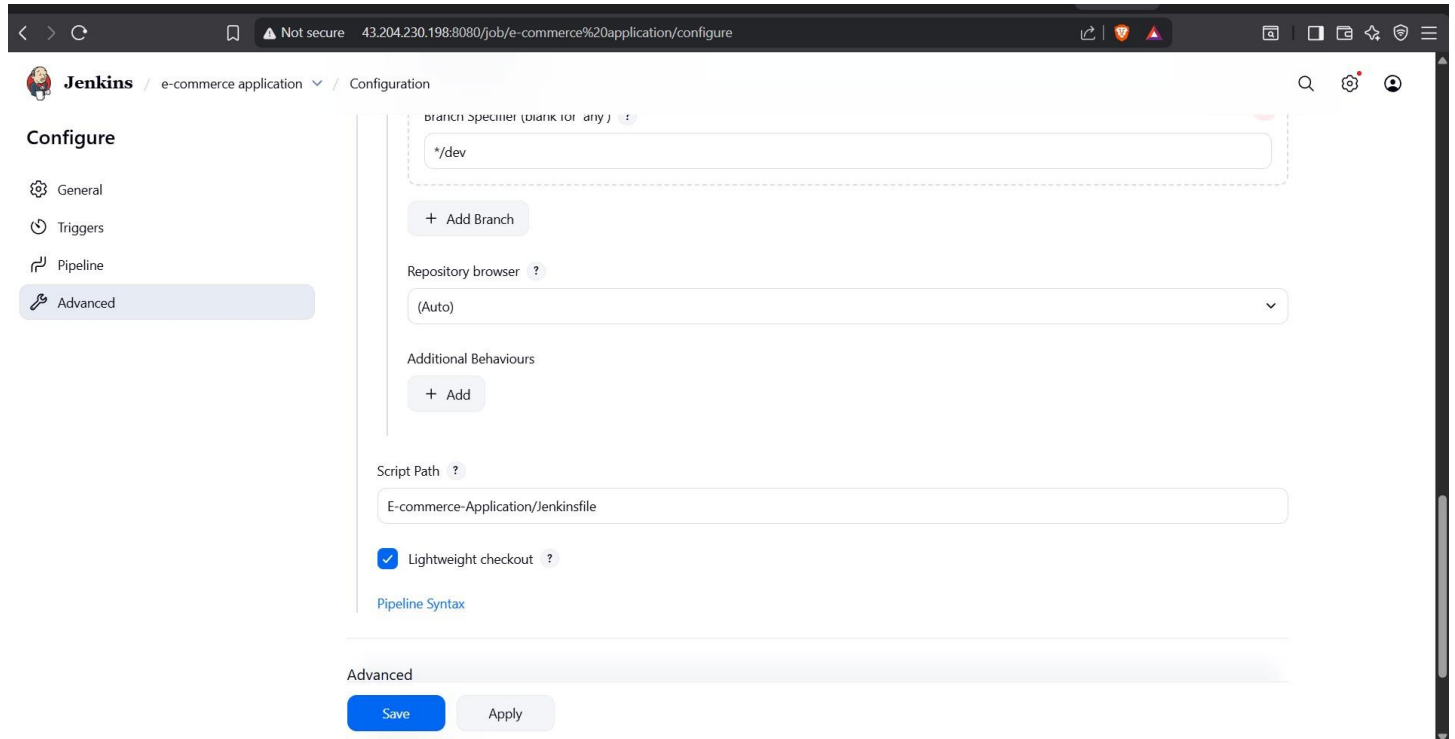
Repositories ?

Repository URL ?

https://github.com/sudharsanps115/FINAL-PROJECT

[Save](#) [Apply](#)

15] JENKINS PIPELINE CONFIG 3



This screenshot shows the Jenkins configuration page for a pipeline named 'e-commerce application'. The 'Advanced' tab is selected in the left sidebar. The configuration includes a 'branch specifier (blank for any)' set to '*/dev', a 'Repository browser' set to '(Auto)', and a 'Script Path' of 'E-commerce-Application/Jenkinsfile'. The 'Lightweight checkout' option is checked. At the bottom, there are 'Save' and 'Apply' buttons.

branch specifier (blank for any) :

+ Add Branch

Repository browser ?

Additional Behaviours
 + Add

Script Path ?

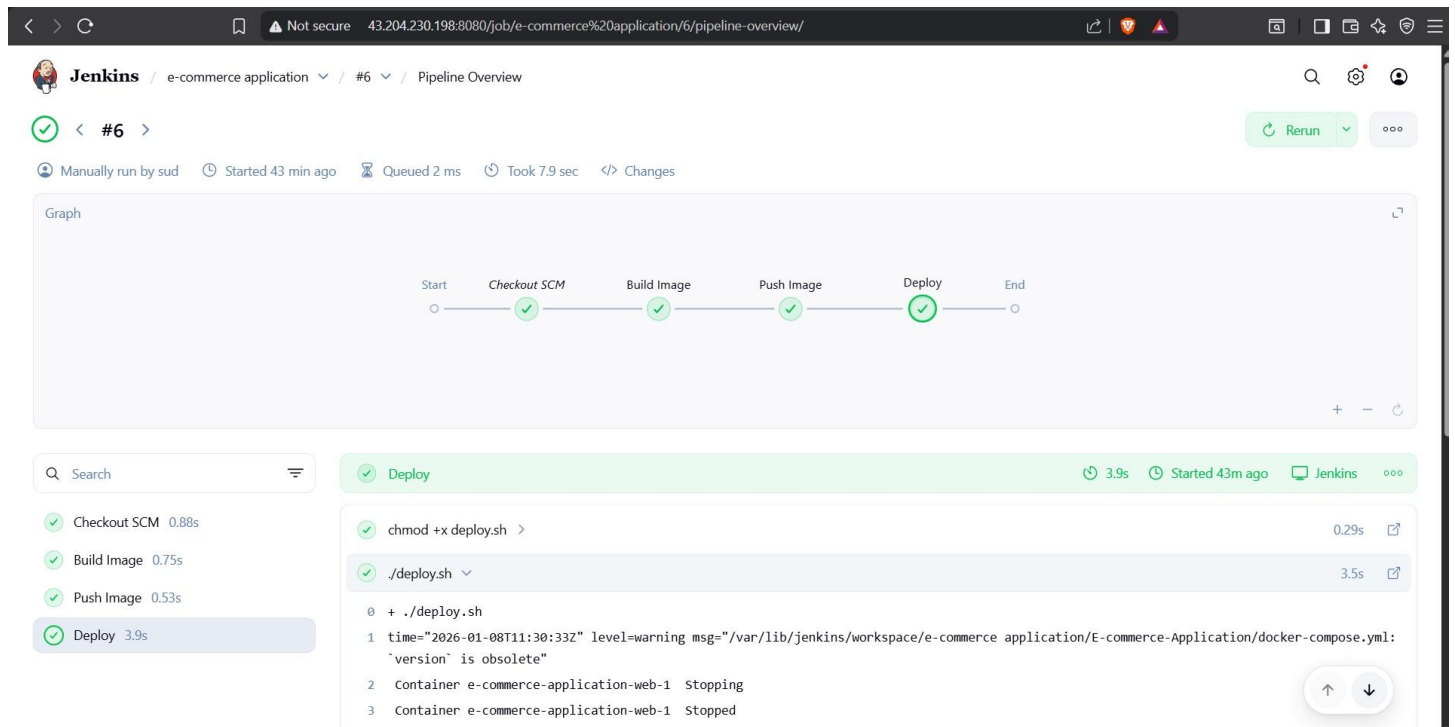
☒ Lightweight checkout ?

[Pipeline Syntax](#)

Advanced

Save Apply

16] JENKINS PIPELINE OVERVIEW



This screenshot shows the Jenkins Pipeline Overview page for the 'e-commerce application' job, specifically for build #6. The pipeline graph shows a successful execution with steps: Start, Checkout SCM, Build Image, Push Image, Deploy, and End. The 'Deploy' step is highlighted in green. Below the graph, a list of steps shows the duration for each: Checkout SCM (0.88s), Build Image (0.75s), Push Image (0.53s), and Deploy (3.9s). The 'Deploy' step is expanded, showing the command 'chmod +x deploy.sh' and the execution of './deploy.sh', which includes a warning about an obsolete 'version' field and the stopping of the 'e-commerce-application-web-1' container.

Manually run by sud Started 43 min ago Queued 2 ms Took 7.9 sec </> Changes

Graph

Start Checkout SCM Build Image Push Image Deploy End

Deploy 3.9s Started 43m ago Jenkins

chmod +x deploy.sh 0.29s

./deploy.sh 3.5s

```
0 + ./deploy.sh
1 time="2026-01-08T11:30:33Z" level=warning msg="/var/lib/jenkins/workspace/e-commerce application/E-commerce-Application/docker-compose.yml:
`version` is obsolete"
2 Container e-commerce-application-web-1 Stopping
3 Container e-commerce-application-web-1 Stopped
```


17] PROJECT- 3 MAIN EC2

Instances (1/2) Info Last updated less than a minute ago

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
project-3	i-0961b7ee60990b1bb	Running	t3.small	3/3 checks passed	View alarms +	ap-south-1b
project-3-depl...	i-0441dc47b0744e66e	Running	t3.micro	Initializing	View alarms +	ap-south-1b

i-0961b7ee60990b1bb (project-3)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

Instance ID i-0961b7ee60990b1bb	Public IPv4 address 43.204.230.198 open address	Private IPv4 addresses 172.31.10.230
IPv6 address -	Instance state Running	Public DNS ec2-43-204-230-198.ap-south-1.compute.amazonaws.com open address

18] PROJECT-3 DEPLOYMENT EC2

Instances (1/2) Info Last updated 1 minute ago

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
project-3	i-0961b7ee60990b1bb	Running	t3.small	3/3 checks passed	View alarms +	ap-south-1b
project-3-depl...	i-0441dc47b0744e66e	Running	t3.micro	Initializing	View alarms +	ap-south-1b

i-0441dc47b0744e66e (project-3-deploying)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

Instance ID i-0441dc47b0744e66e	Public IPv4 address 65.2.74.222 open address	Private IPv4 addresses 172.31.0.160
IPv6 address -	Instance state Running	Public DNS ec2-65-2-74-222.ap-south-1.compute.amazonaws.com open address

19] SECURITY GROUP OF DEPLOYMENT EC2

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type	Protocol	Port range	Source	Description - optional	
sgr-013a56e6a528db99	SSH	TCP	22	Cust...		Delete
sgr-00d763525b02f5701	Custom TCP	TCP	8080	Cust...	49.47.218.116/32	Delete
sgr-079b87d192d795efa	HTTP	TCP	80	Cust...	0.0.0.0/0	Delete
-	Custom TCP	TCP	3000	Any...	0.0.0.0/0	Delete

[Add rule](#)

20] WEEBHOOK TRIGGER SETUP

Webhooks / Manage webhook

[Settings](#) [Recent Deliveries](#)

We'll send a post request to the URL below with details of any subscribed events. You can also specify which data format you'd like to receive (JSON, x-www-form-urlencoded, etc). More information can be found in [our developer documentation](#).

Payload URL *
http://43.204.230.198:8080/github-webhook/

Content type *
application/json

Secret

SSL verification
By default, we verify SSL certificates when delivering payloads.
☒ **Enable SSL verification** ☐ **Disable (not recommended)**

Which events would you like to trigger this webhook?
☒ **Just the push event.**
☐ **Send me everything.**

21] DOCKER , DOCKER COMPOSE SETUP AND LOGIN

```
Go version:      gol.23.1
Git commit:      28.2.2-0ubuntu1~24.04.1
Built:           Wed Sep 10 14:16:39 2025
OS/Arch:         linux/amd64
Context:         default
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Get "http://%2Fvar%2Frun%2Fdocker.sock/v1.50/vers
on": dial unix /var/run/docker.sock: connect: permission denied
ubuntu@ip-172-31-0-160:~$ docker-compose version
command 'docker-compose' not found, but can be installed with:
sudo snap install docker # version 28.4.0, or
sudo apt install docker-compose # version 1.29.2-6
See 'snap info docker' for additional versions.
ubuntu@ip-172-31-0-160:~$ docker login

USING WEB-BASED LOGIN

Info → To sign in with credentials on the command line, use 'docker login -u <username>'

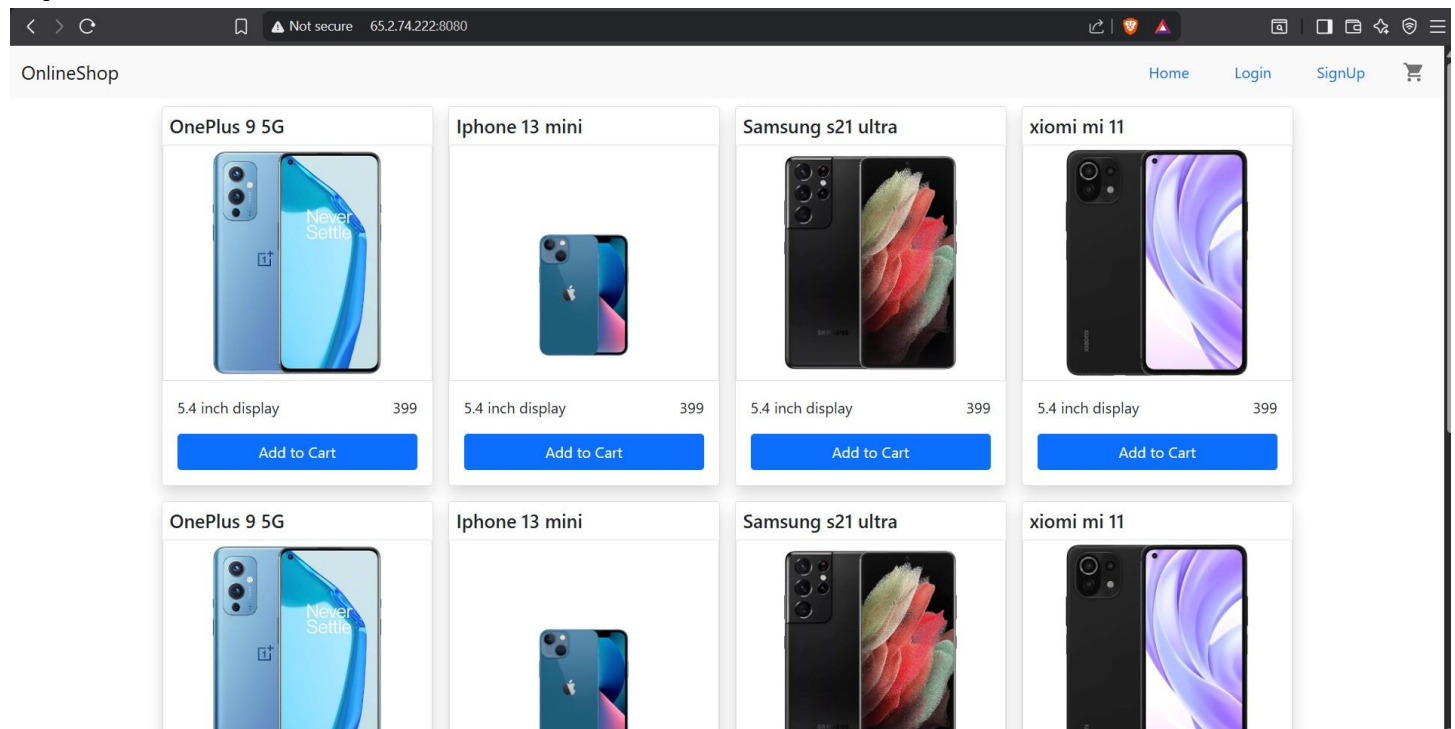
Your one-time device confirmation code is: WRDV-KDFJ
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...

WARNING! Your credentials are stored unencrypted in '/home/ubuntu/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
ubuntu@ip-172-31-0-160:~$
```

22] WEB APP FROM THE JENKINS PIPELINE & HOST THROUGH DEPLOYMENT EC2



23] DOCKER IMG BUILD AND RUN USING DEPLOYMENT EC2

```
ubuntu@ip-172-31-0-160:~$ docker pull sudharsanps115/dev:latest
latest: Pulling from sudharsanps115/dev
1074353eec0d: Pull complete
25f453064fd3: Pull complete
567f84da6fbd: Pull complete
da7c973d8b92: Pull complete
33f95a0f3229: Pull complete
085c5e5aaa8e: Pull complete
0abf9e567266: Pull complete
de54cb821236: Pull complete
76185d7aa564: Pull complete
4624e75b55b6: Pull complete
Digest: sha256:4051d6e148c0972135b8838368ab11e866fb7bceab20d7fbb038552f2c63593
Status: Downloaded newer image for sudharsanps115/dev:latest
docker.io/sudharsanps115/dev:latest
ubuntu@ip-172-31-0-160:~$ docker run -d -p 8080:80 sudharsanps115/dev:latest
89c52f7b320f732f41bd17039f45529edc986fetc3b613c745f9a0630c6d9f8b
ubuntu@ip-172-31-0-160:~$
```

24] PROMETHEUS INSTALLATION

```
ubuntu@ip-172-31-10-230:~$ mkdir grafana-monitoring
ubuntu@ip-172-31-10-230:~$ cd grafana-monitoring/
ubuntu@ip-172-31-10-230:~/grafana-monitoring$ nano prometheus.yml
ubuntu@ip-172-31-10-230:~/grafana-monitoring$ nano Dockerfile
ubuntu@ip-172-31-10-230:~/grafana-monitoring$ docker build -t my-prometheus .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 3.072kB
Step 1/2 : FROM prom/prometheus
latest: Pulling from prom/prometheus
9d85dc8d0609: Pulling fs layer
d0f7326b7716: Pulling fs layer
3dccaafa3f67b: Pulling fs layer
d956c9c5fe9e: Pulling fs layer
1d8e8fd2e272: Pulling fs layer
1dccc9f415d: Pulling fs layer
e5d54fbf8e1: Pulling fs layer
37404d8f503a: Pulling fs layer
b30c77c91326: Pulling fs layer
d76a56e8adff: Pulling fs layer
1dccc9f415d: Waiting
e5d54fbf8e1: Waiting
37404d8f503a: Waiting
b30c77c91326: Waiting
d76a56e8adff: Waiting
d956c9c5fe9e: Waiting
1d8e8fd2e272: Waiting
d0f7326b7716: Download complete
9d85dc8d0609: Verifying Checksum
```

```
< > C ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=ap-south-1&connType=standard&instanceId=i...
aws Search [Alt+S] Asia Pacific (Mumbai) Account ID: 0459-9814-6012 sudharsan

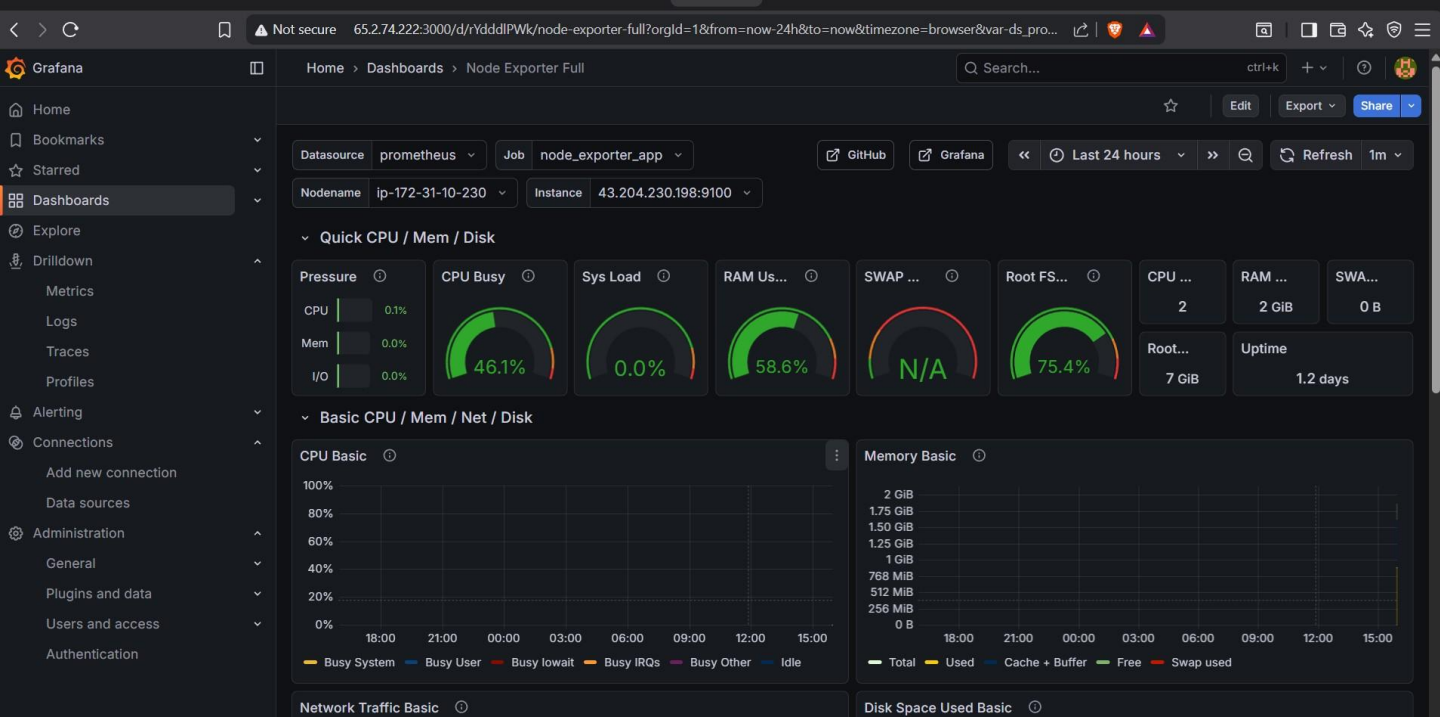
e5d54fbf8ee1: Verifying Checksum
e5d54fbf8ee1: Download complete
37404d8f503a: Verifying Checksum
37404d8f503a: Download complete
b30c77c91326: Verifying Checksum
b30c77c91326: Download complete
d76a56e8adff: Verifying Checksum
d76a56e8adff: Download complete
3dcca3fa3f67b: Pull complete
d956c9c5fe9e: Pull complete
1d8e8fd2e272: Pull complete
1dccc9f415d: Pull complete
e5d54fbf8ee1: Pull complete
37404d8f503a: Pull complete
b30c77c91326: Pull complete
d76a56e8adff: Pull complete
Digest: sha256:1f0f50f06acaceb0f5670d2c8a658a599affe7b0d8e78b898c1035653849a702
Status: Downloaded newer image for prom/prometheus:latest
---> 937690d77350
Step 2/2 : ADD prometheus.yml /etc/prometheus/
---> b340b0f7bb9c
Successfully built b340b0f7bb9c
Successfully tagged my-prometheus:latest
ubuntu@ip-172-31-10-230:~/grafana-monitoring$ docker run -d \
  --name prometheus \
  -p 9090:9090 \
  --restart=always \
  my-prometheus
8245dca402e5a4f046bf530b153c07cbef370ca896c441b9b69b416cab6ef689
ubuntu@ip-172-31-10-230:~/grafana-monitoring$
```

25] GRAFANA INSTALLATION

```
< > C ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=ap-south-1&connType=standard&instanceId=i...
aws Search [Alt+S] Asia Pacific (Mumbai) Account ID: 0459-9814-6012 sudharsan

ubuntu@ip-172-31-10-230:~/grafana-monitoring$ docker run -d \
  --name grafana \
  -p 3000:3000 \
  --restart=always \
  grafana/grafana
Unable to find image 'grafana/grafana:latest' locally
latest: Pulling from grafana/grafana
014e56e61396: Pull complete
dafcd853cdea: Pull complete
fe0fb9ad3b64: Pull complete
f50da64eedfd: Pull complete
4a965f51fa32: Pull complete
835a6d0e4902: Pull complete
a2a0d06388d6: Pull complete
a7d3fdc2e5af: Pull complete
c8585378023b: Pull complete
d2772eb63bbd: Pull complete
Digest: sha256:2175aaa91c96733d86d31cf270d5310b278654b03f5718c59de12a865380a31f
Status: Downloaded newer image for grafana/grafana:latest
41c5ff05657d9d8c28e754effd679395f66da074d3ee7b55948910ff29e1dbc8
ubuntu@ip-172-31-10-230:~/grafana-monitoring$ docker ps
CONTAINER ID   IMAGE               COMMAND                  CREATED        STATUS        PORTS                               NAME
41c5ff05657d   grafana/grafana    "/run.sh"               4 seconds ago Up 4 seconds  0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp   grafana
8245dca402e5   my-prometheus      "/bin/prometheus --c..." 6 minutes ago Up 6 minutes  0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp   prometheus
3230593f85b2   devops-build:late /docker-entrypoint..." About an hour ago Up About an hour 0.0.0.0:80->80/tcp, [::]:80->80/tcp           e-commerce-application-web-1
ubuntu@ip-172-31-10-230:~/grafana-monitoring$
```


26] MONITORING HEALTH



27] PROMETHEUS DASHBOARD

